Question: Is there an anomaly? If so, of which type and where?

Ground truth: **Hypoapnea – start: 04:12 | end : 04:20**

Recursive Language Model (RLM)

*The question appears to be related to a **sleep recording**. Let me verify the time-series length, number of channels and channel names.* `<CODE>`

`<REPL output appended to context chain>`

The signal spans 2 minutes of recording and 13 channels. Sleep anomalies are usually characterized by sudden respiratory obstructions. I will call a subagent to identify anomalous candidates on the four most relevant channels to capture these anomalies: FLOW, PRES, THO and ABD. `<CODE>`

Sub-RVLM

Let me investigate the signal further... `<CODE>` *On the four channels, I will calculate rolling mean, deviation and sudden drops and perform a spectral decomposition of the signal...* `<CODE>`

`Answer: Potential respiratory anomaly between 04:12 and 04:20`

*Good findings, now call a **second subagent** to classify the respiratory anomaly at that precise location for these channels.* `<CODE>`

`Answer: Hypoapnea - start: 04:12 | end : 04:20`

REPL Environment

Python execution runtime

```
>> sub_rlm (signal = signal[:200],
channels = {'FLOW', 'PRES', 'THO',
'ABD'}, question = q)

>> for b in rolling_windows:
      m, d = measure_stats(b)
      print(f"Mean {m}, dev: {d}")

>> plt.plot([x], y, data=s[10:400])
>> look(plt.show())
```

Environment Variables

`signal` ⟶ time-series data

`sub_rlm()` ⟶ callable rlm

`look()` ⟶ view plot

`FINAL()` ⟶ submit answer

Packages

```
import numpy
import scipy
import matplotlib
```

Image-mode add. package

